INCH-POUND

MIL-H-87111/2E
13 September 1993
SUPERSEDING
MIL-H-87111/2D
18 May 1990

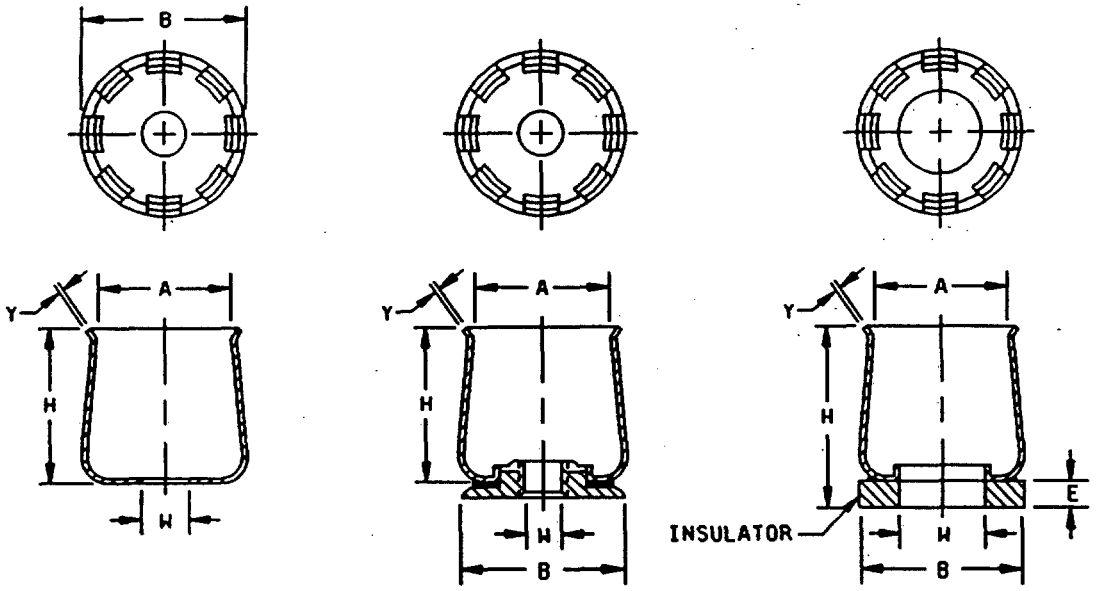
MILITARY SPECIFICATION SHEET

HEAT SINKS, SEMICONDUCTOR DEVICES, MATERIAL CLASSES 2, 3, AND 5 (RETAINER CLIP TYPE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-H-87111.

Part or Identifying Numbers (PIN's) M87111/02-3*07 through M87111/02-3*15 are inactive for new design as of 26 December 1988.



CONFIGURATION A CONFIGURATION B CONFIGURATION C

FIGURE 1. Heat sinks (retainer clip type).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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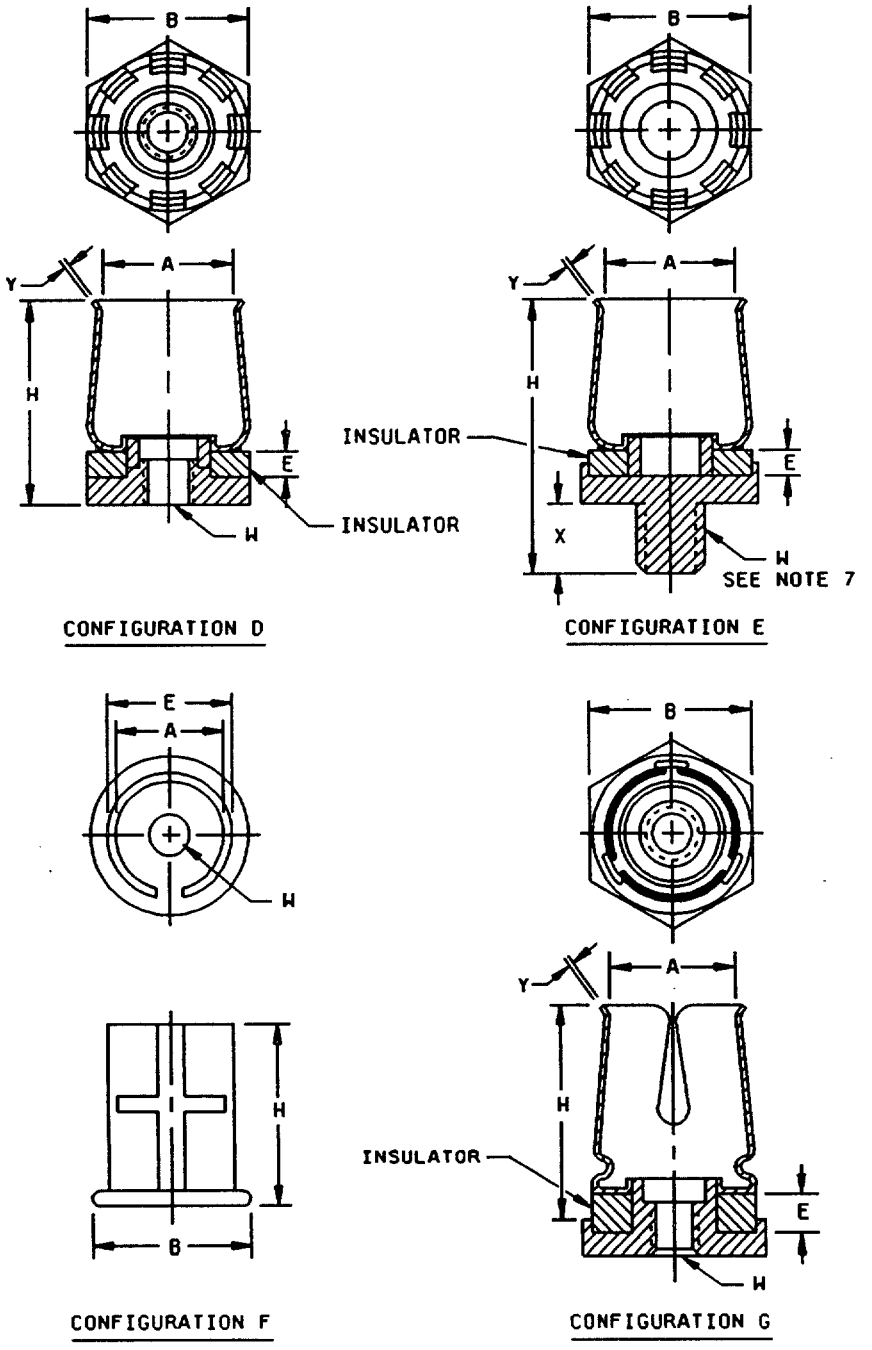
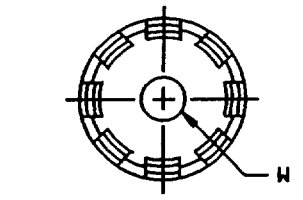
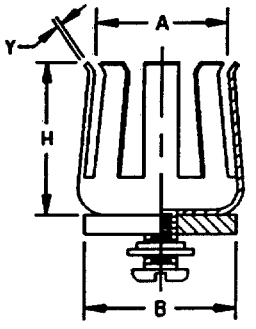


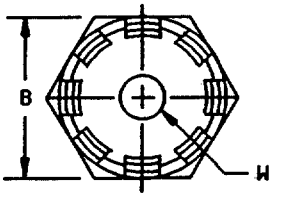
FIGURE 1. Heat sinks (retainer clip type) - Continued.

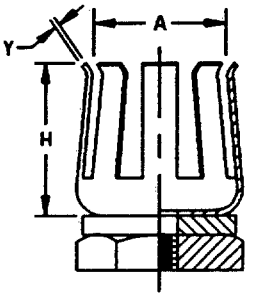
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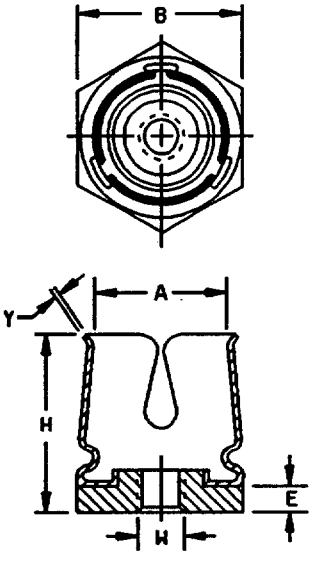


CONFIGURATION I

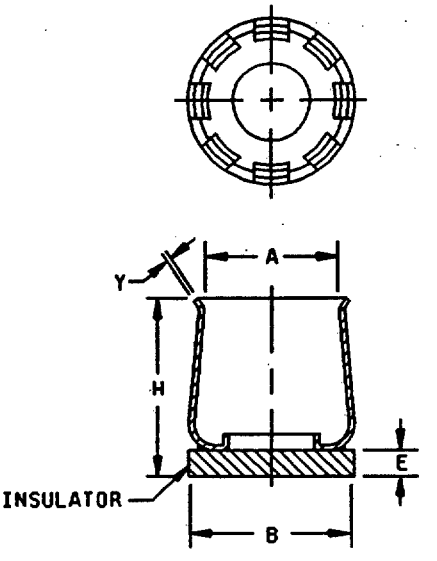




CONFIGURATION J



CONFIGURATION K



CONFIGURATION L

FIGURE 1. Heat sinks (retainer clip type) - Continued.

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Dash number (see note 6)	JEDEC case size	Config-uration	Dimensions (see notes)						
			A Dia +.005 (0.13)	B	E	H	W	X	Y (see note 8)
2*01	TO-18	A	.170 -.010 (4.32 - 0.25)	.250 ±.015 (6.35 ±0.38)	N/A	.219 ±.015 (5.56 ±0.38)	.067 +.004 -.001 (1.70 +0.10 -0.03)	N/A	.012 (0.30)
2*02	TO-5	A	.295 -.015 (7.49 -0.38)	.375 ±.010 (9.52 ±0.25)	N/A	.320 ±.035 (8.13 ±0.89)	.098 +.015 -.001 (2.49 +0.38 -0.03)	N/A	.013 (.33)
2*03	TO-8	A	.470 -.015 (11.94 -0.38)	.545 ±.015 (13.84 ±0.38)	N/A	.281 ±.015 (7.14 ±0.38)	.130 (3.30)	N/A	.010 (.25)
2*04	TO-18	B	.170 -.010 (4.32 -0.25)	.235 ±.025 (5.97 ±0.64)	N/A	.290 ±.025 (7.37 ±0.64)	.086-56 UNC-2B	N/A	.012 (.31)
2*05	TO-5	B	.295 -.015 (7.49 -0.38)	.375 ±.015 (9.52 ±.038)	N/A	.370 ±.035 (9.40 ±0.38)	.112-40 UNC-2B	N/A	.013 (.33)
2*06	TO-8	B	.475 -.015 (12.07 -0.38)	.545 ±.015 (13.84 ±0.38)	N/A	.370 ±.015 (9.40 ±0.38)	.112-40 UNC-2B	N/A	.010 (.25)
3*07	TO-18	C	.176 -.015 (4.47 -0.38)	.220 ±.015 (5.59 ±0.38)	.060 (1.52)	.281 (7.14)	.067 (1.70)	N/A	.012 (.31)
3*08	TO-5	C	.305 -.015 (7.75 -0.38)	.375 ±.015 (9.52 ±0.38)	.050 ±.010 (1.27 ±0.25)	.400 ±.015 (10.16 ±0.38)	.185 ±.005 (4.70 ±0.13)	N/A	.013 (.33)
3*09	TO-8	C	.470 -.015 (11.94 -0.38)	.545 ±.015 (13.84 ±0.38)	.070 (1.78)	.281 ±.015 (7.14 ±0.38)	.130 (3.30)	N/A	.010 (.25)
3*10	TO-8	D	.470 -.015 (11.94 -0.38)	.545 ±.015 (13.84 ±0.38)	.070 (1.78)	.281 ±.015 (7.14 ±0.38)	.112-40-UNC-2A	N/A	.010 (.25)
3*11	TO-18	E	.170 -.010 (4.32 -0.25)	.250 ±.010 (6.35 ±0.25)	.060 (1.52)	.550 ±.015 (13.97 ±0.38)	.112-40-UNC-2A	.190 (4.83)	.012 (.31)
3*12	TO-5	E	.295 -.015 (7.49 -0.38)	.375 ±.010 (9.52 ±0.25)	.060 (1.52)	.656 +.012 -.050 (16.66 +0.30 -1.27)	.138-32 UNC-2A	.190 (4.83)	.013 (.33)

FIGURE 1. Heat sinks (retainer clip type) - Continued.

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Dash number (see note 6)	JEDEC case size	Config-uration	Dimensions (see notes)						
			A Dia +.005 (0.13)	B	E	H	W	X	Y (see note 8)
3*13	TO-5	F	.315 -.006 (8.00 -0.15)	.375 ±.007 (9.52 ±0.18)	.351 (8.92)	.400 ±.016 (10.16 ±0.41)	.112-40-UNC-2B	N/A	N/A
3*14	TO-18	G	.170 -.010 (4.32 -0.25)	.250 ±.015 (6.35 ±0.38)	.050 (1.27)	.350 ±.015 (8.89 ±0.38)	.086-56-UNC-2B	N/A	.010 (0.25)
3*15	TO-5	G	.295 -.015 (7.49 -0.38)	.375 ±.015 (9.52 ±0.38)	.050 (1.27)	.445 ±.015 (11.30 ±0.38)	.112-40-UNC-2B	N/A	.010 (0.25)
2*16	TO-5	I	.295 -.015 (7.49 -0.38)	.375 ±.015 (9.52 ±0.38)	N/A	.370 ±.015 (9.40 ±0.38)	.112-40-UNC-2B	.156 (3.96)	.013 (0.33)
2*17	TO-5	J	.295 -.015 (7.49 -0.38)	.375 ±.015 (9.52 ±0.38)	N/A	.410 ±.015 (10.41 ±0.38)	.112-40-UNC-2B	N/A	.013 (0.33)
5*18	TO-18	J	.170 -.010 (4.32 -0.25)	.250 ±.015 (6.35 ±0.38)	N/A	.290 ±.015 (7.37 ±0.38)	.086-56-UNC-2B	N/A	.010 (0.25)
5*19	TO-18	E	.170 -.010 (4.32 -0.25)	.250 ±.010 (6.35 ±0.25)	N/A	.495 ±.015 (12.57 ±0.38)	.086-56-UNC-2A	.187 (4.75)	.010 (0.25)
5*20	TO-5	E	.295 -.010 (7.49 -0.25)	.375 ±.015 (9.52 ±0.38)	N/A	.585 ±.020 (14.86 ±0.46)	.138-32-UNC-2A	.187 (4.75)	.010 (0.25)
2*21	TO-18	I	.170 -.010 (4.32 -0.25)	.215 ±.015 (5.46 ±0.38)	N/A	.370 ±.015 (9.40 ±0.38)	.086-56-UNC-2B	.125 (3.17)	.012 (0.30)
2*22	TO-8	I	.470 -.015 (11.94 -0.38)	.550 ±.015 (13.97 ±0.38)	N/A	.370 ±.015 (9.40 ±0.38)	.112-40-UNC-2B	.156 (3.96)	.010 (0.25)
5*23	TO-18	K	.170 -.010 (4.32 -0.25)	.250 ±.015 (6.35 ±0.38)	.050 (1.27)	.295 ±.015 (7.49 ±0.38)	.086-56-UNC-2B	N/A	.010 (0.25)
5*24	TO-5	K	.295 -.010 (7.49 -0.25)	.375 ±.015 (9.52 ±0.38)	.090 (2.29)	.400 ±.015 (10.16 ±0.38)	.112-40-UNC-2B	N/A	.010 (0.25)
5*25	TO-8	K	.470 -.010 (11.94 -0.25)	.550 ±.015 (13.97 ±0.38)	N/A	.375 ±.015 (9.52 ±0.38)	.112-40-UNC-2B	.156 (3.96)	N/A
5*26	TO-18	B	.170 -.010 (4.32 -0.25)	.215 ±.015 (5.46 ±0.38)	N/A	.330 ±.015 (8.38 ±0.38)	.086-56-UNC-2B	N/A	.010 (0.25)
5*27	TO-5	B	.295 -.010 (7.49 -0.25)	.365 ±.015 (9.27 ±0.38)	N/A	.365 ±.010 (9.27 ±0.38)	.112-40-UNC-2B	N/A	.010 (0.25)
2*28	TO-18	E	.170 -.010 (4.32 -0.25)	.250 ±.010 (6.35 ±0.25)	N/A	.495 ±.020 (12.57 ±0.51)	.138-32-UNC-2A	.187 (4.75)	.012 (0.30)
2*29	TO-18	E	.170 -.010 (4.32 -0.25)	.250 ±.010 (6.35 ±0.25)	N/A	.495 ±.020 (12.57 ±0.51)	.112-40-UNC-2A	.187 (4.75)	.012 (0.30)
3*30	TO-18	L	.176 -.015 (4.47 -0.38)	.220 ±.015 (5.59 ±0.38)	.060 (1.52)	.281 (7.14)	N/A	N/A	.012 (0.31)
3*31	TO-5	L	.305 -.015 (7.75 -0.38)	.375 ±.015 (9.52 ±0.38)	.060 (1.52)	.385 ±.015 (9.78 ±0.38)	N/A	N/A	.015 (0.38)

FIGURE 1. Heat sinks (retainer clip type) - Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses and are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. Drawings depict general design characteristics. Shape of actual parts may vary slightly within dimensions shown.
5. Number of fingers may vary depending on manufacturer.
6. The asterisk in the dash number represents finish (i.e., B, C, D, G, or N).
7. .0625 inch (1.588 mm) undercut to minimum thread diameter at top portion of threaded stud (configuration E only).
8. Dimension Y value is before application of finish.

FIGURE 1. Heat sinks (retainer clip type) - Continued.

REQUIREMENTS:

Dimensions and configurations: See figure 1.

Material options (see MIL-H-87111 for reference specifications):

Beryllium copper (class 2).

Beryllium copper with beryllium oxide insulator (class 3).

Beryllium copper with epoxy insulator (class 5).

CAUTION: Class 3 devices contain beryllium oxide insulators. Heat sinks of this type are not hazardous in their finished state; however, grinding, sandblasting, machining, sectioning, or similar operations should not be performed on these parts unless proper safety precautions are observed. These operations may produce beryllium dust or fumes which, if inhaled, could cause serious pulmonary disease. Inhalation of beryllium particles in excess of OSHA occupational exposure standards should be avoided. Class 3 parts may be recycled or disposed of according to the manufacturer's instructions.

Finish options (see MIL-H-87111 for reference specifications):

Black chemical/black ebanol (finish B).

Insulating coating (finish C).

Cadmium plating (finish D).

Gold plating (finish G).

Nickel plating (finish N).

Mounting surface flatness: Mounting surfaces shall be flat within .003 inch (0.08 mm) per inch of Total Indicator Reading (TIR).

Surface roughness: Not applicable.

JEDEC case sizes: See figure 1.

Thermal resistance: See table I.

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TABLE I. Thermal resistance. 1/

	TO-5	TO-8	TO-18
Noninsulated; base and cup soldered:	12°C/Watt	5°C/Watt	22.5°C/Watt
Insulated; 2/ base and cup epoxy bonded:	18°C/Watt (60 Hz/500 V dc	N/A	25°C/Watt (60 Hz/250 V dc)
Capacitance at 175 kHz	30 pF		16 pF
Insulated; 2/ beryllium oxide wafer bonded between base and cup:	16°C/Watt (60 Hz/1000 V dc)	N/A	27.5°C/Watt (60 Hz/750 V dc)
Capacitance at 175 kHz	3 to 5.5 pF		1.4 to 2.5 pF

1/ Case to sink; thermal joint compound used on all interfaces. Applies to infinite or constant temperature heat sinks.
2/ Add 10°C/Watt for insulated coated cup clips.

PIN: M87111/02- (plus dash number from figure 1). The material class number immediately follows the dash. The asterisk shall be replaced by a finish letter. PIN example: M87111/02-2B01.

Supersession data: See table II.

TABLE II. Supersession data.

Superseding PIN in accordance with MIL-H-87111/2C 26 December 1988	Superseded PIN 1/ in accordance with MIL-H-87111/2B 18 April 1988
M87111/02-5C26	M87111/02-3*07
M87111/02-5C27	M87111/02-3*08
M87111/02-5C25	M87111/02-3*09
M87111/02-5C25	M87111/02-3*10
M87111/02-5C19	M87111/02-3*11
M87111/02-5C20	M87111/02-3*12
M87111/02-5C24	M87111/02-3*13
M87111/02-5C23	M87111/02-3*14
M87111/02-5C24	M87111/02-3*15

1/ The asterisk in the PIN represents the finish (i.e., B, C, D, G, or N).

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - TD
Air Force - 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5999-0309)

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